1. The basic circuit

The simplest circuit is a battery, some connecting wire and a resistor in a loop. The battery creates a potential difference (voltage) that pushes electrons through the wire. Current flows out of the positive terminal, through the resistor, and back to the battery. MARKER-CIRCUIT-P1.

2. The water-hose analogy

Think of a garden water hose. Battery voltage is the water-supply pressure: higher pressure pushes harder. The outlet valve is the resistor: closing it reduces the flow. The water flow itself is the electric current. Increase resistance and flow drops; open the valve (less resistance) and flow rises. MARKER-HOSE-P2.

3. Ohm’s law and the triangle

Ohm’s law links the three quantities: V = I x R. Draw it as a triangle with V on top and I, R below. Cover the value you want: cover V and you read I multiplied by R; cover I and you read V divided by R; cover R and you read V divided by I. MARKER-TRIANGLE-P3.

4. Units and symbols

In engineering voltage is denoted V and measured in volts. Resistance is denoted R (or the omega sign) and measured in ohms. Current is denoted I and measured in amps. So V=IR reads volts = amps x ohms. MARKER-UNITS-P4.

5. Worked example: 12V across 250 ohms

Take a 12 volt battery with a 250 ohm resistor across it. Cover I in the triangle: I = V / R. 12 volts divided by 250 ohms gives 0.048 amps, which is 48 milliamps - all without using a meter. Connecting wire has a small resistance too, but here the total is treated as 250 ohms. MARKER-EXAMPLE-P5.

6. Series, parallel and the board practical

An ammeter joins the loop in series so the full current passes through it. A voltmeter connects in parallel across the resistor only: red (positive) where current enters, black (negative) where it leaves. A rheostat in series varies the current for the experiment. Red terminals are positive, black are negative, and every connection must be tight. MARKER-PRACTICAL-P6.